KAIZEN MATHS

Lesson starter

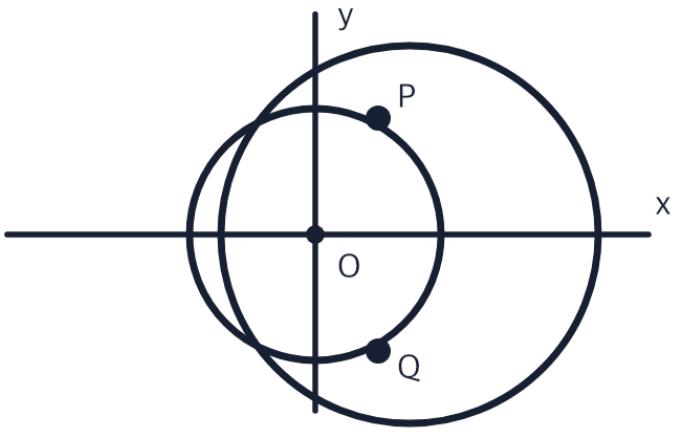
Polar Coordinates and Polar Curves

Intersections and Gradients

Name _____

SA1

Find the intersections of the polar curves.



Find the intersections of $r = 5$ and $r = 10\cos\theta$.

ALG-294F-L3 · SA

Show your working

KAIZEN MATHS

Lesson starter

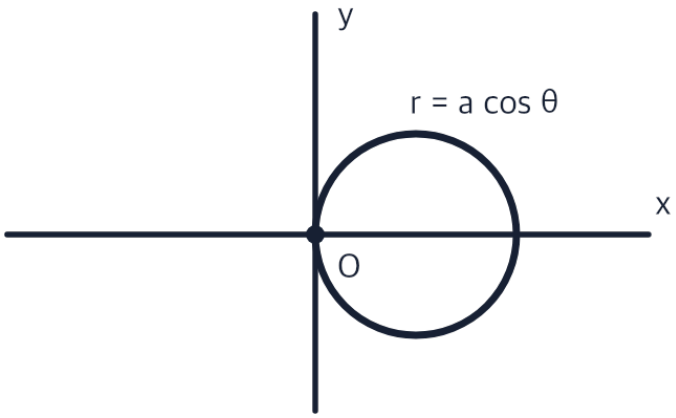
Polar Coordinates and Polar Curves

Intersections and Gradients

Name _____

SB1

Find where the curve meets the given ray.



The curve is $r = 6\cos\theta$. Find the point where it meets the ray $\theta = \pi/2$.

ALG-294F-L3 · SB

Show your working

KAIZEN MATHS

Lesson starter

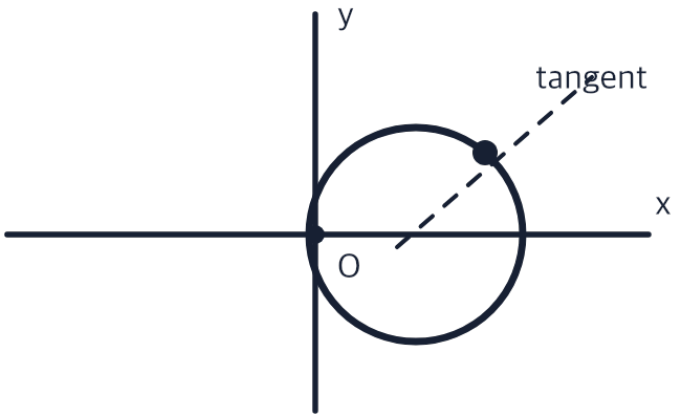
Polar Coordinates and Polar Curves

Intersections and Gradients

Name _____

SC1

Find dy/dx at the given value of θ.



For the polar curve $r = 4\cos\theta$, find dy/dx at $\theta = \pi/3$.

Show your working

KAIZEN MATHS

Lesson starter

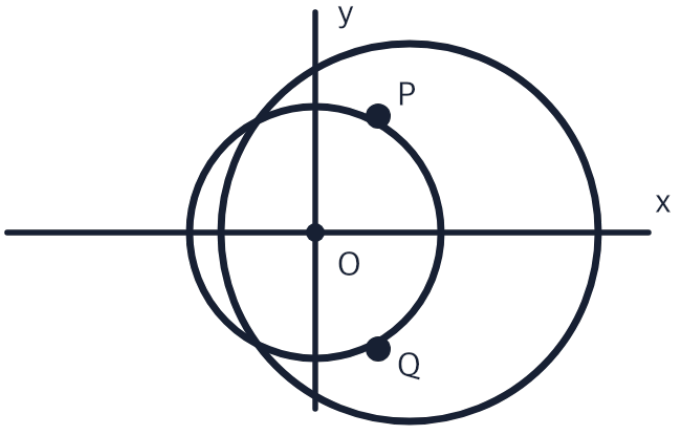
Polar Coordinates and Polar Curves

Intersections and Gradients

Name _____

SD1

Find the intersections of the polar curves.



Find the intersections of $r = 3$ and $r = 6\cos\theta$.

Show your working

KAIZEN MATHS

Exit ticket

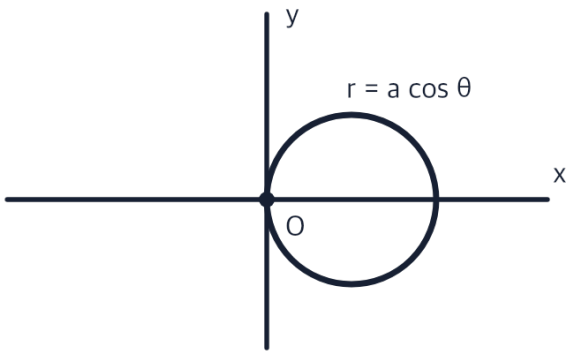
Polar Coordinates and Polar Curves

Intersections and Gradients

Name

EA1

Find where the curve meets the given ray.



The curve is r = 5 cos θ. Find the point where it meets the ray θ = π/3.

Show your working

KAIZEN MATHS

Exit ticket

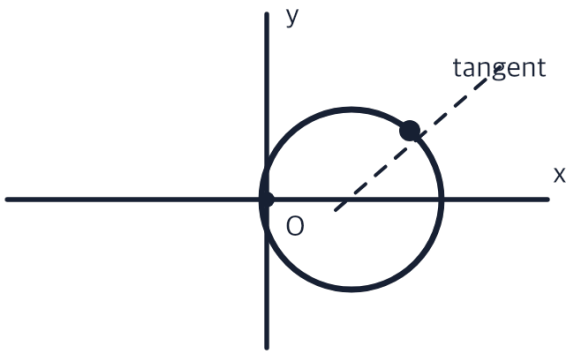
Polar Coordinates and Polar Curves

Intersections and Gradients

Name

EB1

Find dy/dx at the given value of θ.



For the polar curve r = 4 cos θ, find dy/dx at θ = π/4.

Show your working

KAIZEN MATHS

Exit ticket

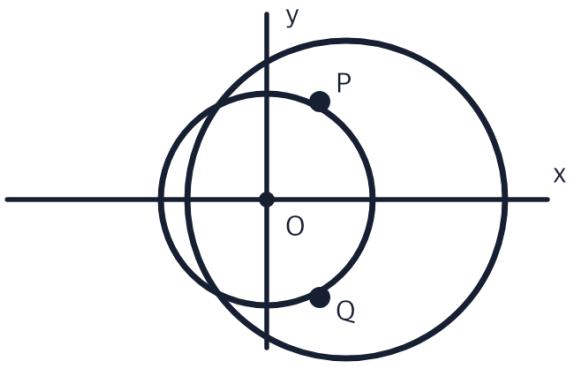
Polar Coordinates and Polar Curves

Intersections and Gradients

Name

EC1

Find the intersections of the polar curves.



Find the intersections of r = 2 and r = 4 cos θ.

Show your working

KAIZEN MATHS

Exit ticket

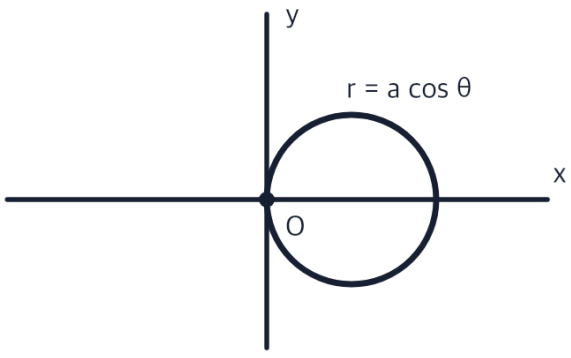
Polar Coordinates and Polar Curves

Intersections and Gradients

Name

EA1

Find where the curve meets the given ray.



The curve is r = 5 cos θ. Find the point where it meets the ray θ = π/3.

Show your working

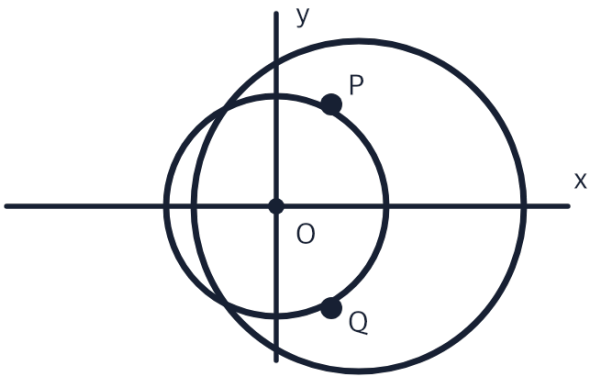
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

ALG-294F-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Find the intersections of the polar curves.



Find the intersections of r = 5 and r = 10 cos θ.

Answer:

θ = π/3 or 5π/3, with r = 5

Working:

Set the two expressions for r equal.

$5 = 10\cos\theta$.

$\cos\theta = \frac{5}{10} = \frac{1}{2}$.

For $0 \leq \theta < 2\pi$, $\theta = \pi/3$ or $5\pi/3$.

At both intersections, r = 5.

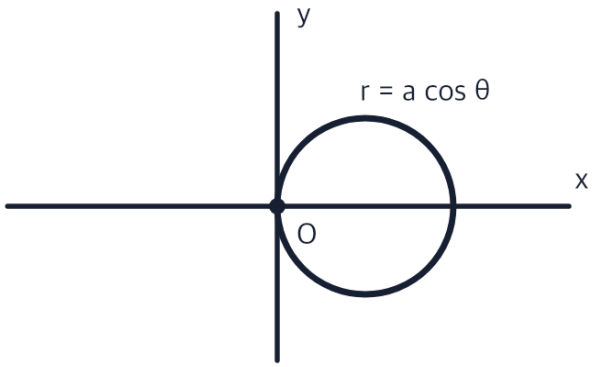
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

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SB1

Find where the curve meets the given ray.



The curve is r = 6 cos θ. Find the point where it meets the ray θ = π/2.

Answer:

r = 0, so the polar point is (0, π/2)

Working:

Substitute θ = π/2 into r = 6 cos θ.

$r = 6\cos(\pi/2)$.

$\cos(\pi/2) = 0$.

So $r = 0$.

The polar point is (0, π/2).

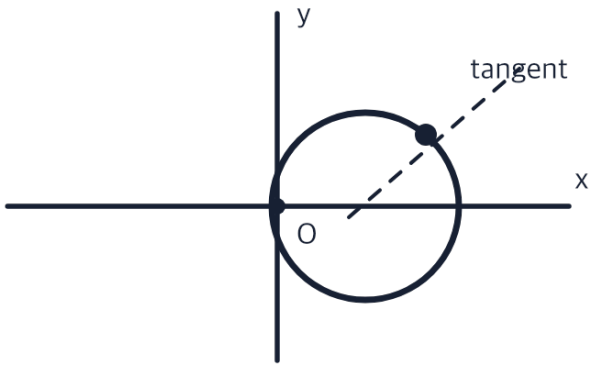
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

ALG-294F-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Find dy/dx at the given value of θ.



For the polar curve r = 4 cos θ, find dy/dx at θ = π/3.

Answer:

dy/dx = √3/3

Working:

Use $\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$.

For r = 4 cos θ, r' = -4 sin θ.

Substitute r and r' into the formula.

$\frac{dy}{dx} = \frac{(-4\sin\theta)\sin\theta + (4\cos\theta)\cos\theta}{(-4\sin\theta)\cos\theta - (4\cos\theta)\sin\theta}$.

$\frac{dy}{dx} = \frac{4(\cos^2\theta - \sin^2\theta)}{-8\sin\theta\cos\theta}$ = -cot(2θ).

At θ = π/3, -cot(2π/3) = √3/3.

Therefore dy/dx = √3/3.

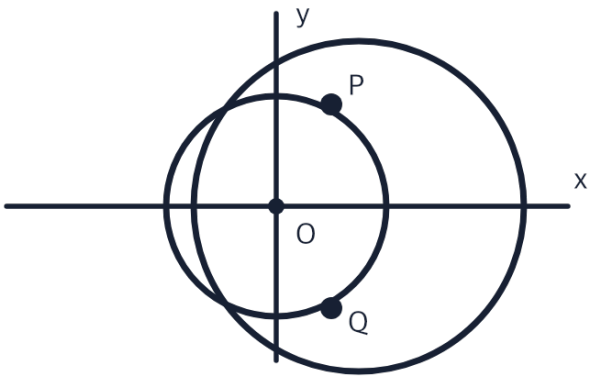
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

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SD1

Find the intersections of the polar curves.



Find the intersections of r = 3 and r = 6 cos θ.

Answer:

θ = π/3 or 5π/3, with r = 3

Working:

Set the two expressions for r equal.

$3 = 6\cos\theta$.

$\cos\theta = \frac{3}{6} = \frac{1}{2}$.

For $0 \leq \theta < 2\pi$, $\theta = \pi/3$ or $5\pi/3$.

At both intersections, r = 3.

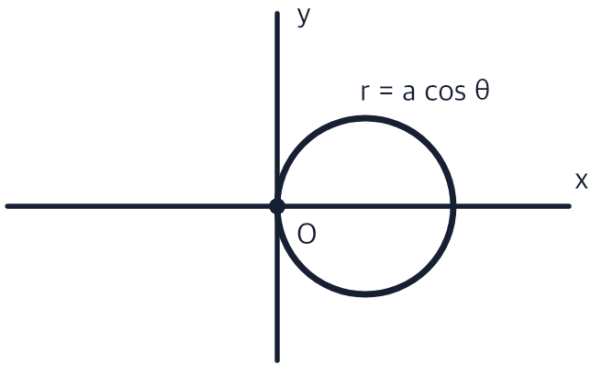
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

ALG-294F-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Find where the curve meets the given ray.



The curve is r = 5 cos θ. Find the point where it meets the ray θ = π/3.

Answer:

r = $\frac{5}{2}$, so the polar point is ($\frac{5}{2}$, π/3)

Working:

Substitute θ = π/3 into r = 5 cos θ.

$r = 5\cos(\pi/3)$.

$\cos(\pi/3) = \frac{1}{2}$.

So $r = \frac{5}{2}$.

The polar point is ($\frac{5}{2}$, π/3).

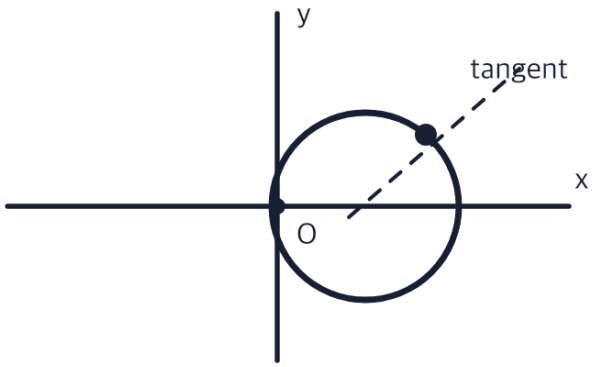
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

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EB1

Find dy/dx at the given value of θ.



For the polar curve r = 4 cos θ, find dy/dx at θ = π/4.

Answer:

dy/dx = 0

Working:

Use $\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$.

For r = 4 cos θ, r' = -4 sin θ.

Substitute r and r' into the formula.

$\frac{dy}{dx} = \frac{(-4\sin\theta)\sin\theta + (4\cos\theta)\cos\theta}{(-4\sin\theta)\cos\theta - (4\cos\theta)\sin\theta}$.

$\frac{dy}{dx} = \frac{4(\cos^2\theta - \sin^2\theta)}{-8\sin\theta\cos\theta}$ = -cot(2θ).

At θ = π/4, -cot(π/2) = 0.

Therefore dy/dx = 0.

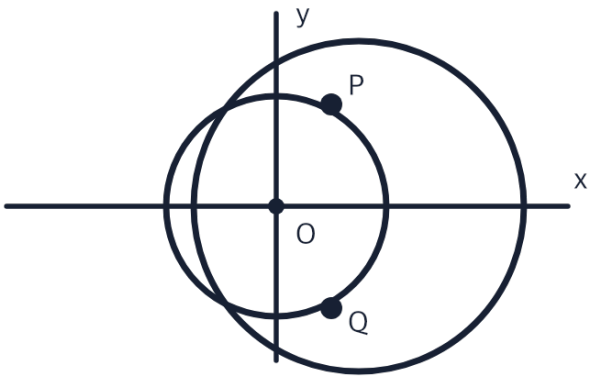
Teacher answers & working

Polar Coordinates and Polar Curves · Intersections and Gradients

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EC1

Find the intersections of the polar curves.



Find the intersections of r = 2 and r = 4 cos θ.

Answer:

θ = π/3 or 5π/3, with r = 2

Working:

Set the two expressions for r equal.

$2 = 4\cos\theta$.

$\cos\theta = \frac{2}{4} = \frac{1}{2}$.

For $0 \leq \theta < 2\pi$, $\theta = \pi/3$ or $5\pi/3$.

At both intersections, r = 2.